

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	Hard

Question

The cost of renting a large canopy tent for up to 10 days is \$430 for the first day and \$215 for each additional day. Which of the following equations gives the cost y , in dollars, of renting the tent for x days, where x is a positive integer and $x \leq 10$?

Answer

- A. $y = 215x + 215$
- B. $y = 430x - 215$
- C. $y = 430x + 215$
- D. $y = 215x + 430$

Correct Answer: A

Rationale

Choice A is correct. It’s given that the cost of renting a large canopy tent is \$430 for the first day and \$215 for each additional day for up to 10 days. For x days of renting the tent, the cost includes \$430 for the first day and \$215 for each of the $(x - 1)$ additional days. It follows that the cost y , in dollars, of renting the tent can be expressed as $y = 430 + 215(x - 1)$, which is equivalent to $y = 430 + 215x - 215$, or $y = 215x + 215$. Therefore, the equation $y = 215x + 215$ gives the cost of renting the tent for x days, where x is a positive integer and $x \leq 10$.

Choice B is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$215, not \$430, and the cost for each additional day is \$430, not \$215.

Choice C is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$645, not \$430, and the cost for each additional day is \$430, not \$215.

Choice D is incorrect. This equation represents a situation where the cost of renting the tent for the first day is \$645, not \$430.